

Statistics for Social Science Research

Lecture 5

30 August 2026

Topics for today

- Introduction to Hypothesis Testing (Chapter 8, Statistics for the Behavioral Sciences)
- Introduction to the t Statistics (Chapter 9, Statistics for the Behavioral Sciences)

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

Chapter 8

Introduction to Hypothesis Testing

8.1 Hypothesis Testing Logic

- Hypothesis testing is one of the most commonly used inferential procedures
- Definition: a statistical method that uses sample data to evaluate the validity of a hypothesis about a population parameter

Logic of Hypothesis Test

- State hypothesis about a population
- Predict the expected characteristics of the sample based on the hypothesis
- Obtain a random sample from the population
- Compare the obtained sample data with the prediction made from the hypothesis
 - If consistent, hypothesis is reasonable
 - If discrepant, hypothesis is rejected

Figure 8.1

Basic Experimental Design

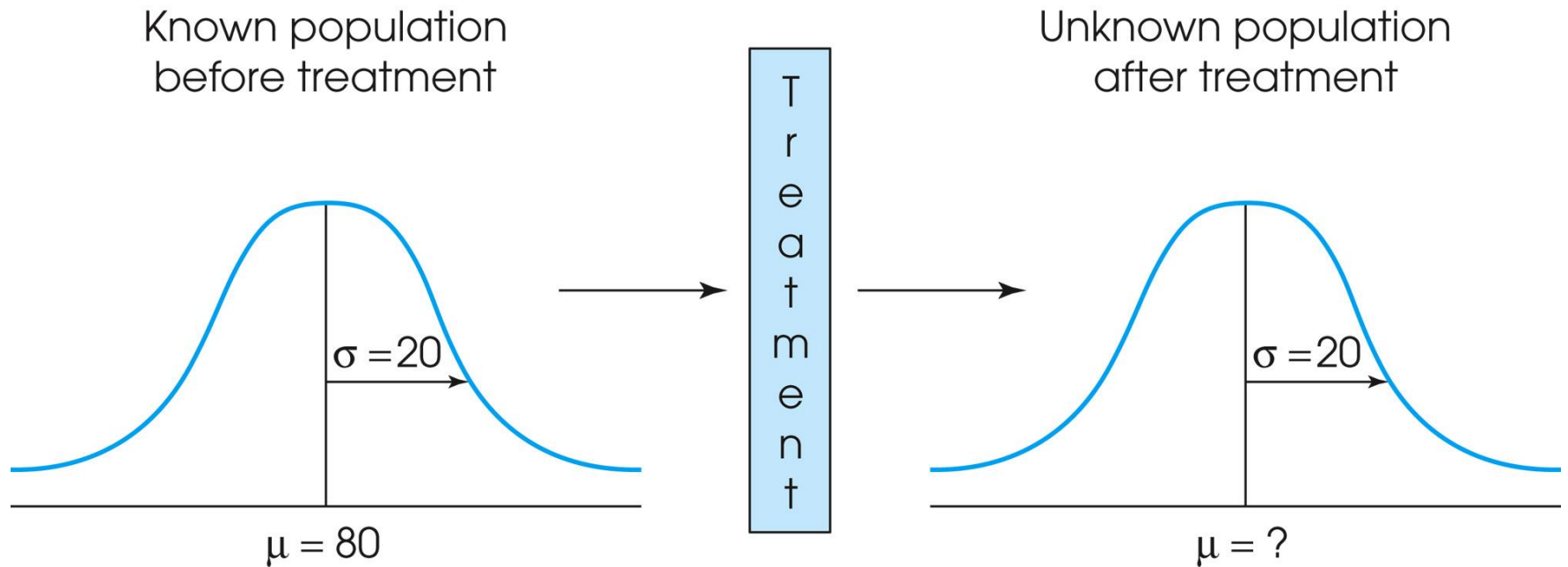
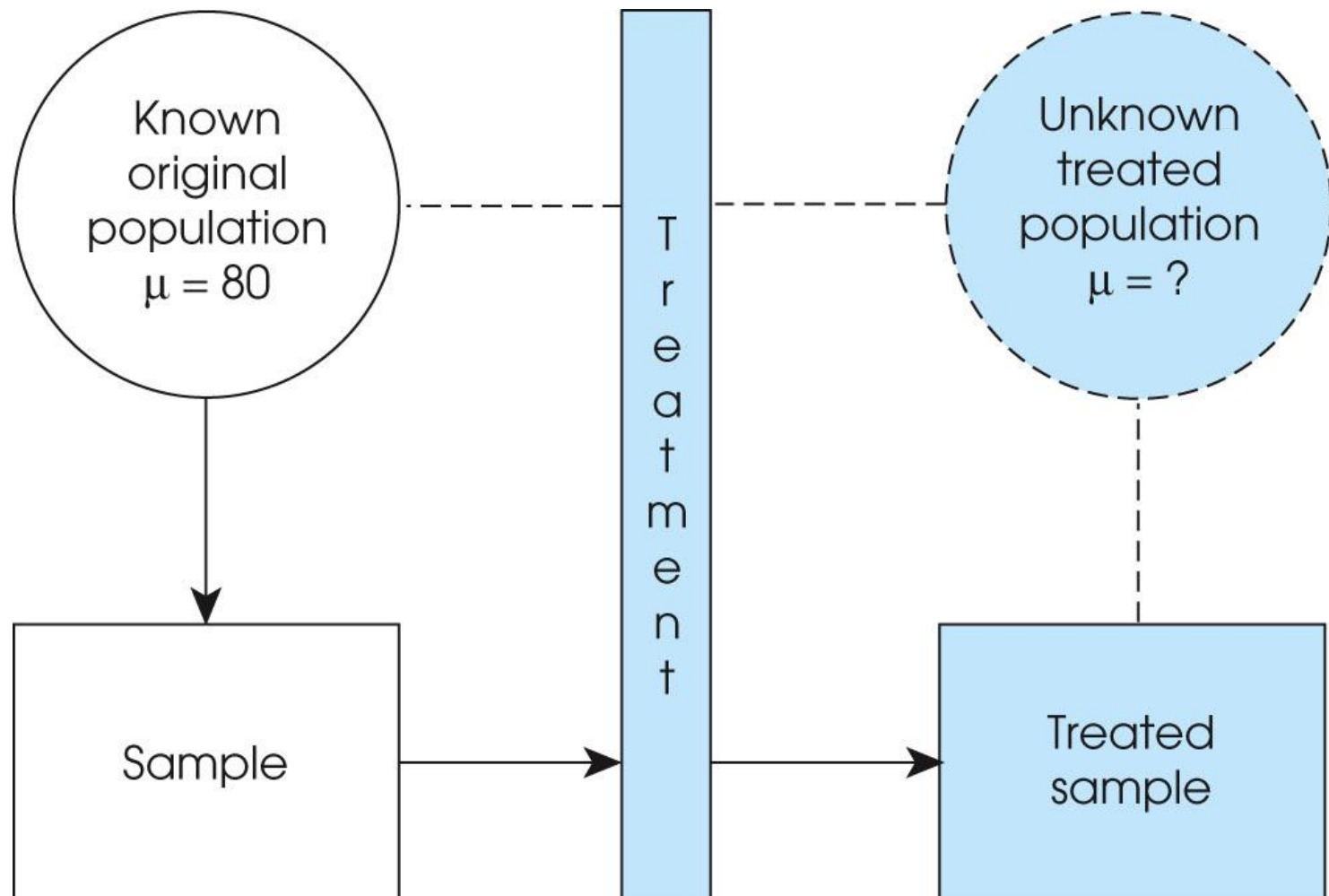


Figure 8.2 Unknown Population in Basic Experimental Design



Four Steps in Hypothesis Testing

Step 1: State the hypotheses

Step 2: Set the criteria for a decision

Step 3: Collect data; compute sample statistics

Step 4: Make a decision

Step 1: State Hypotheses

- Null hypothesis (H_0) states that, in the general population, there is no change, no difference, or is no relationship
- Alternative hypothesis (H_1) states that there is a change, a difference, or there is a relationship in the general population

Step 2: Set the Decision Criterion

- Distribution of sample outcomes is divided
 - Those likely if H_0 is true
 - Those “very unlikely” if H_0 is true
- Alpha level, or significance level, is a probability value used to define “very unlikely” outcomes
- Critical region(s) consist of the extreme sample outcomes that are “very unlikely”
- Boundaries of critical region(s) are determined by the probability set by the alpha level

Figure 8.3 Note “Unlikely” Parts of Distribution of Sample Means

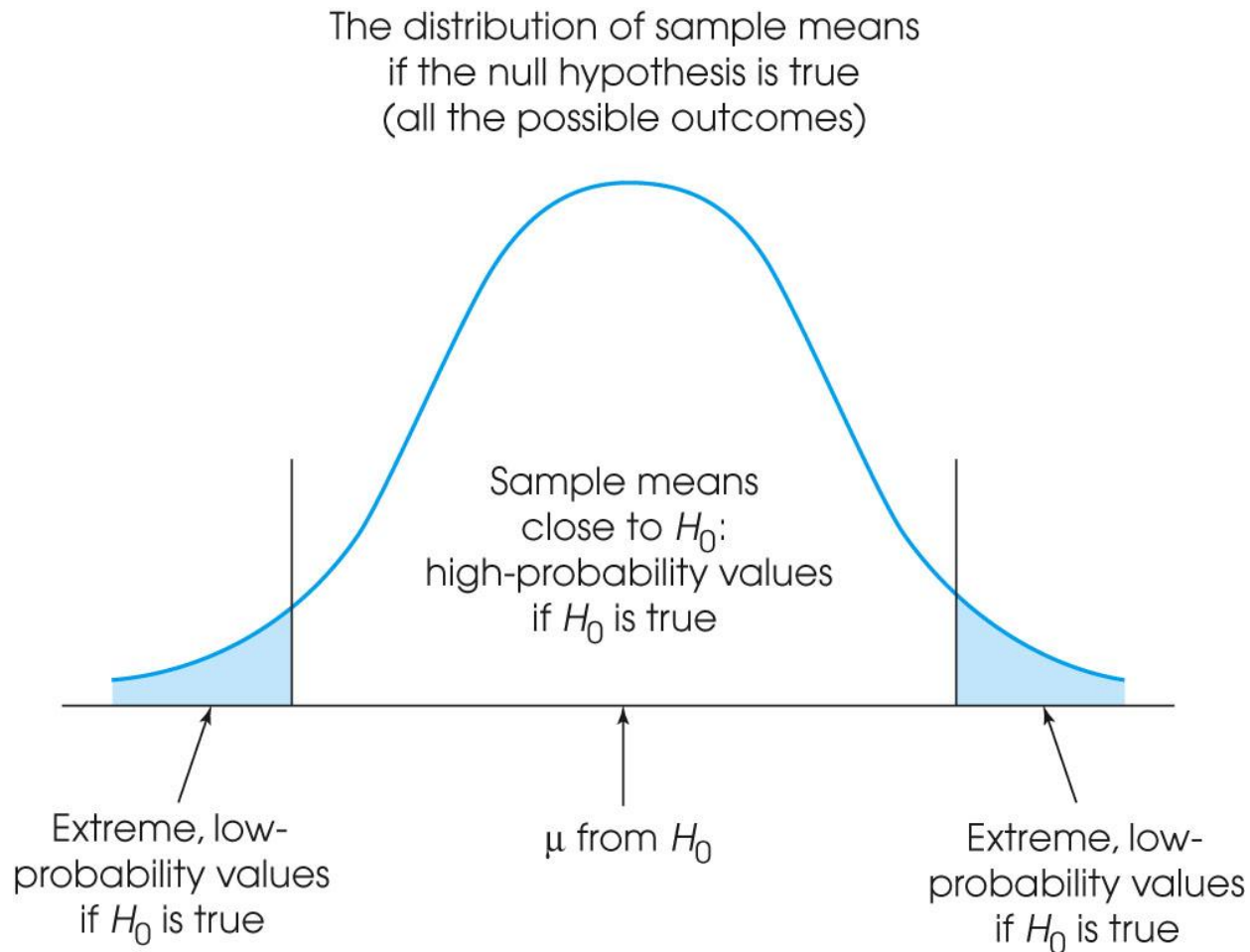
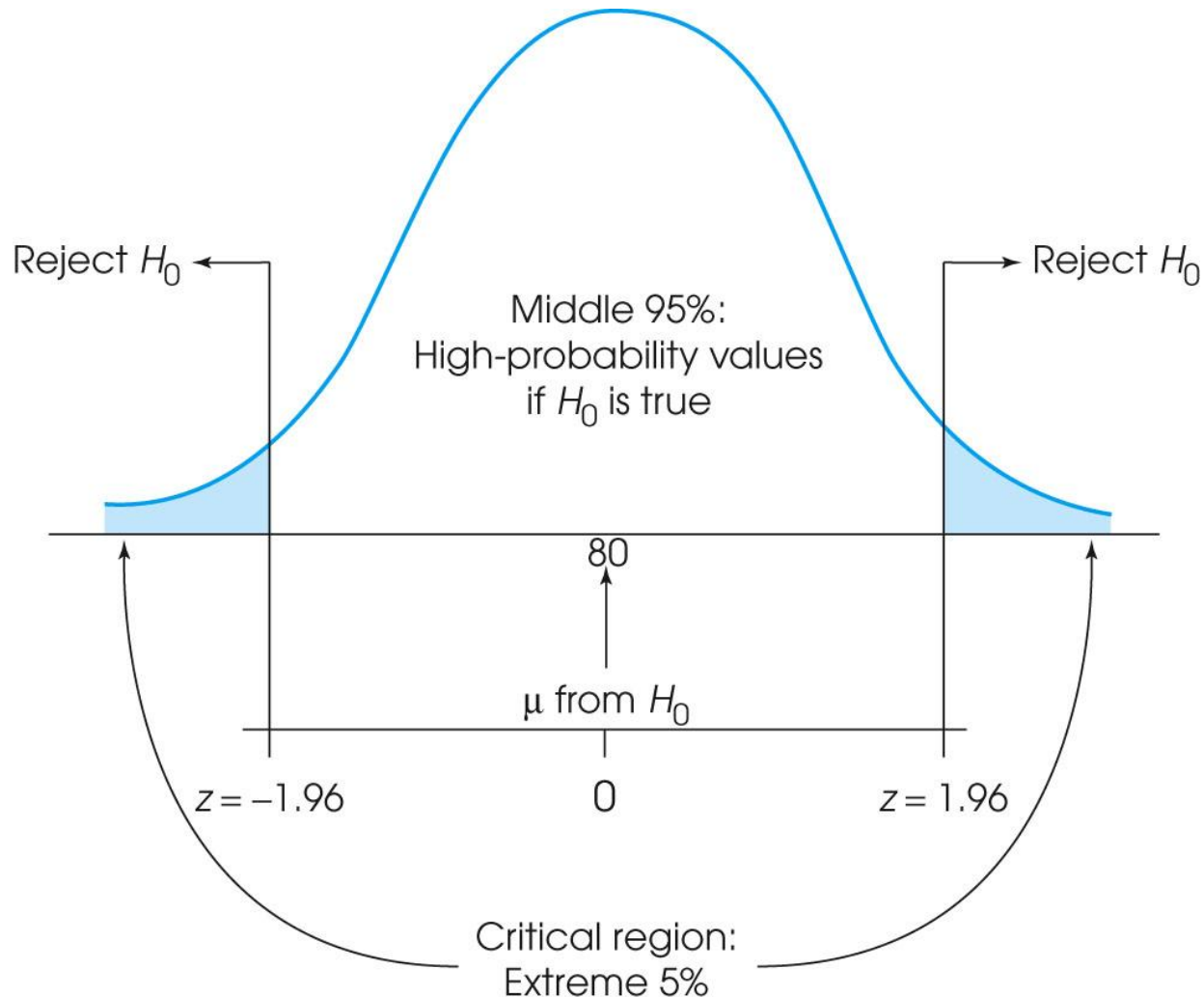


Figure 8.4

Critical region(s) for $\alpha = .05$



Learning Check

- A sports coach is investigating the impact of a new training method. In words, what would the null hypothesis say?

A

- The new training program produces different results from the existing one

B

- The new training program produces results about like the existing one

C

- The new training program produces better results than the existing one

D

- There is no way to predict the results of the new training program

Learning Check - Answer

- A sports coach is investigating the impact of a new training method. In words, what would the null hypothesis say?

A

• The new training program produces different results from the existing one

B

• The new training program produces results about like the existing one

C

• The new training program produces better results than the existing one

D

• There is no way to predict the results of the new training program

Learning Check

- Decide if each of the following statements is True or False.

T/F

- If the alpha level is decreased, the size of the critical region decreases

T/F

- The critical region defines unlikely values if the null hypothesis is true

Learning Check - Answers

True

- Alpha is the proportion of the area in the critical region(s)

True

- This is the definition of “unlikely”

Step 3: Collect Data (and...)

- Data always collected after hypotheses stated
- Data always collected after establishing decision criteria
- This sequence assures objectivity

Step 3: (continued)...

Compute Sample Statistics

- Compute a sample statistic (z-score) to show the exact position of the sample

$$z = \frac{M - \mu}{\sigma_M}$$

- In words, z is the difference between the observed sample mean and the hypothesized population mean divided by the standard error of the mean

Step 4: Make a decision

- If sample statistic (z) is located in the critical region, the null hypothesis is rejected
- If the sample statistic (z) is not located in the critical region, the researcher fails to reject the null hypothesis

Jury Trial:

Hypothesis Testing Analogy

- Trial begins with the null hypothesis “not guilty” (defendant’s innocent plea)
- Police and prosecutor gather evidence (data) relevant to the validity of the innocent plea
- With sufficient evidence against, jury rejects null hypothesis innocence claim to conclude “guilty”
- With insufficient evidence against, jury fails to convict, i.e., fails to reject the “not guilty” claim (but does not conclude defendant is innocent)

Learning Check

- Decide if each of the following statements is True or False.

T/F

- When the z-score is quite extreme, it shows the null hypothesis is true

T/F

- A decision to retain the null hypothesis means you proved that the treatment has no effect

Learning Check - Answer

False

- An extreme z-score is in the critical region—very unlikely if H_0 is true

False

- Failing to reject H_0 does not prove it true; there is just not enough evidence to reject it

8.2 Uncertainty and Errors in Hypothesis Testing

- Hypothesis testing is an inferential process
 - Uses limited information from a sample to make a statistical decision, and then from it a general conclusion
 - Sample data used to make the statistical decision allows us to make an inference and draw a conclusion about a population
- Errors are possible

Type I Errors

- Researcher rejects a null hypothesis that is actually true
- Researcher concludes that a treatment has an effect when it has none
- Alpha level is the probability that a test will lead to a Type I error

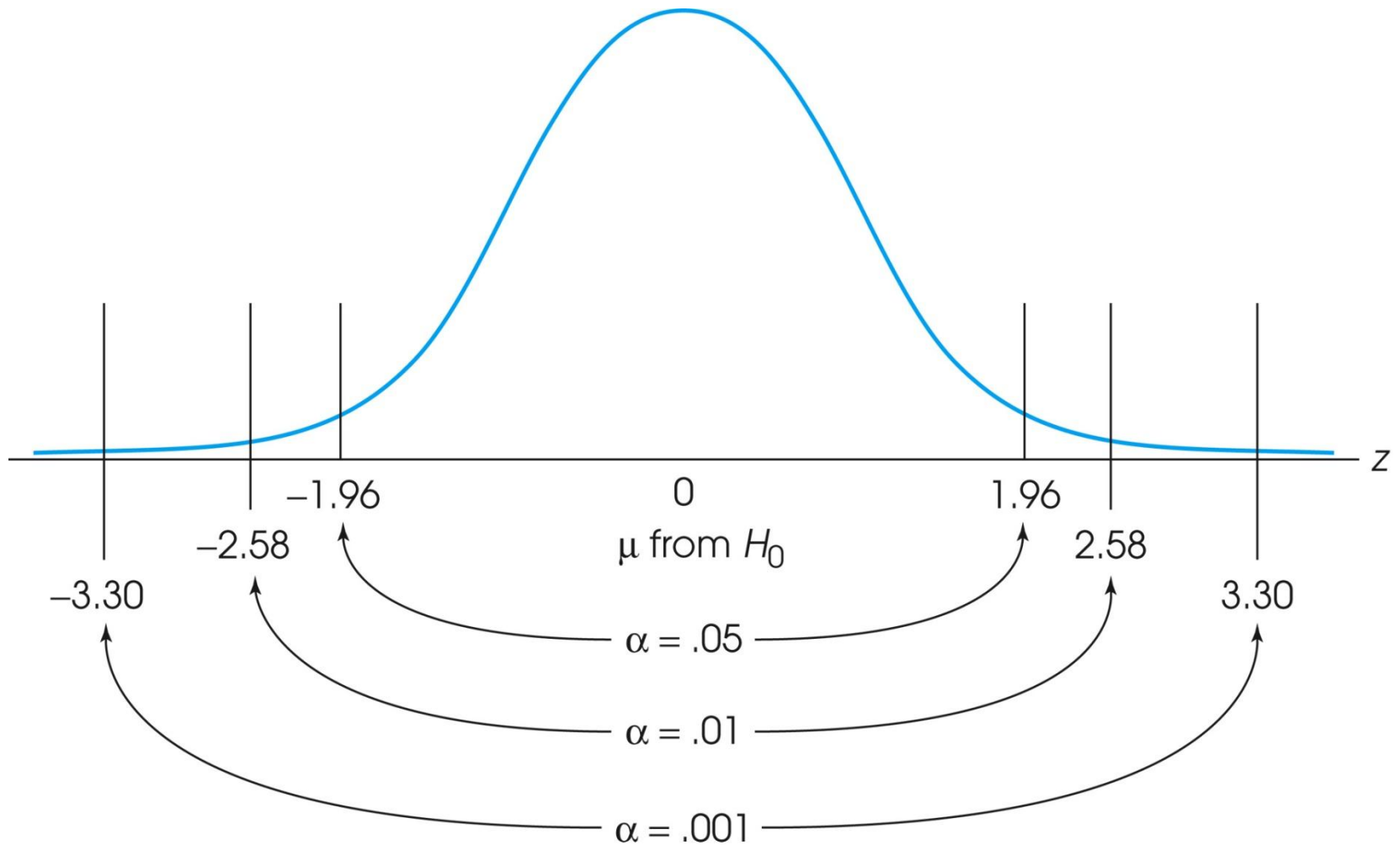
Type II Errors

- Researcher fails to reject a null hypothesis that is really false
- Researcher has failed to detect a real treatment effect
- Type II error probability is not easily identified

Table 8.1

		Actual Situation	
		<i>No Effect = H_0 True</i>	<i>Effect Exists = H_0 False</i>
Researcher's Decision	<i>Reject H_0</i>	Type I error (α)	Decision correct
	<i>Fail to reject H_0</i>	Decision correct	Type II error (β)

Figure 8.5 Location of Critical Region Boundaries



Learning Check

- Decide if each of the following statements is True or False.

T/F

- A Type I error is like convicting an innocent person in a jury trial

T/F

- A Type II error is like convicting a guilty person in a jury trial

Learning Check - Answer

True

- Innocence is the “null hypothesis” for a jury trial; conviction is like rejecting that hypothesis

False

- Convicting a guilty person is not an error; but acquitting a guilty person would be like Type II error

8.3 Hypothesis Testing Summary

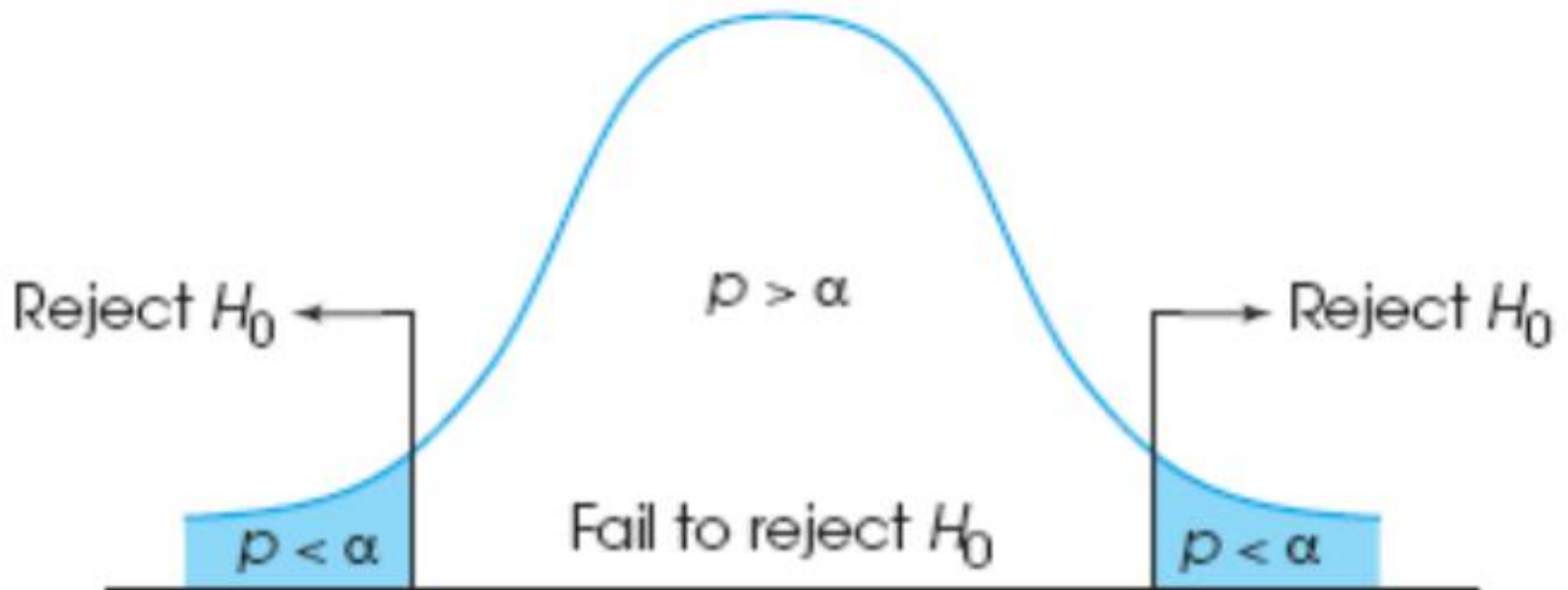
- Step 1: State hypotheses and select alpha level
- Step 2: Locate the critical region
- Step 3: Collect data; compute the test statistic
- Step 4: Make a probability-based decision about H_0 : Reject H_0 if the test statistic is unlikely when H_0 is true—called a “significant” or “statistically significant” result

In the Literature

- A result is significant or statistically significant if it is very unlikely to occur when the null hypothesis is true; conclusion: reject H_0
- In APA format
 - Report that you found a significant effect
 - Report value of test statistic
 - Report the p -value of your test statistic

Figure 8.6

Critical Region for Standard Test



8.3 Assumptions for Hypothesis Tests with z-Scores

- Random sampling
- Independent Observation
- Value of σ is not changed by the treatment
- Normally distributed sampling distribution

Factors that Influence the Outcome of a Hypothesis Test

- Size of difference between sample mean and original population mean
 - Larger discrepancies $\rightarrow$ larger z-scores
- Variability of the scores
 - More variability $\rightarrow$ larger standard error
- Number of scores in the sample
 - Larger n $\rightarrow$ smaller standard error

Learning Check

- A researcher uses a hypothesis test to evaluate $H_0: \mu = 80$. Which combination of factors is most likely to result in rejecting the null hypothesis?

A

• $\sigma = 5$ and $n = 25$

B

• $\sigma = 5$ and $n = 50$

C

• $\sigma = 10$ and $n = 25$

D

• $\sigma = 10$ and $n = 50$

Learning Check - Answer

- A researcher uses a hypothesis test to evaluate $H_0: \mu = 80$. Which combination of factors is most likely to result in rejecting the null hypothesis?

A

• $\sigma = 5$ and $n = 25$

B

• $\sigma = 5$ and $n = 50$

C

• $\sigma = 10$ and $n = 25$

D

• $\sigma = 10$ and $n = 50$

Learning Check

- Decide if each of the following statements is True or False.

T/F

- An effect that exists is more likely to be detected if n is large

T/F

- An effect that exists is less likely to be detected if σ is large

Learning Check - Answers

True

- A larger sample produces a smaller standard error and larger z

True

- A larger standard deviation increases the standard error and produces a smaller z

8.4 Directional Hypothesis Tests

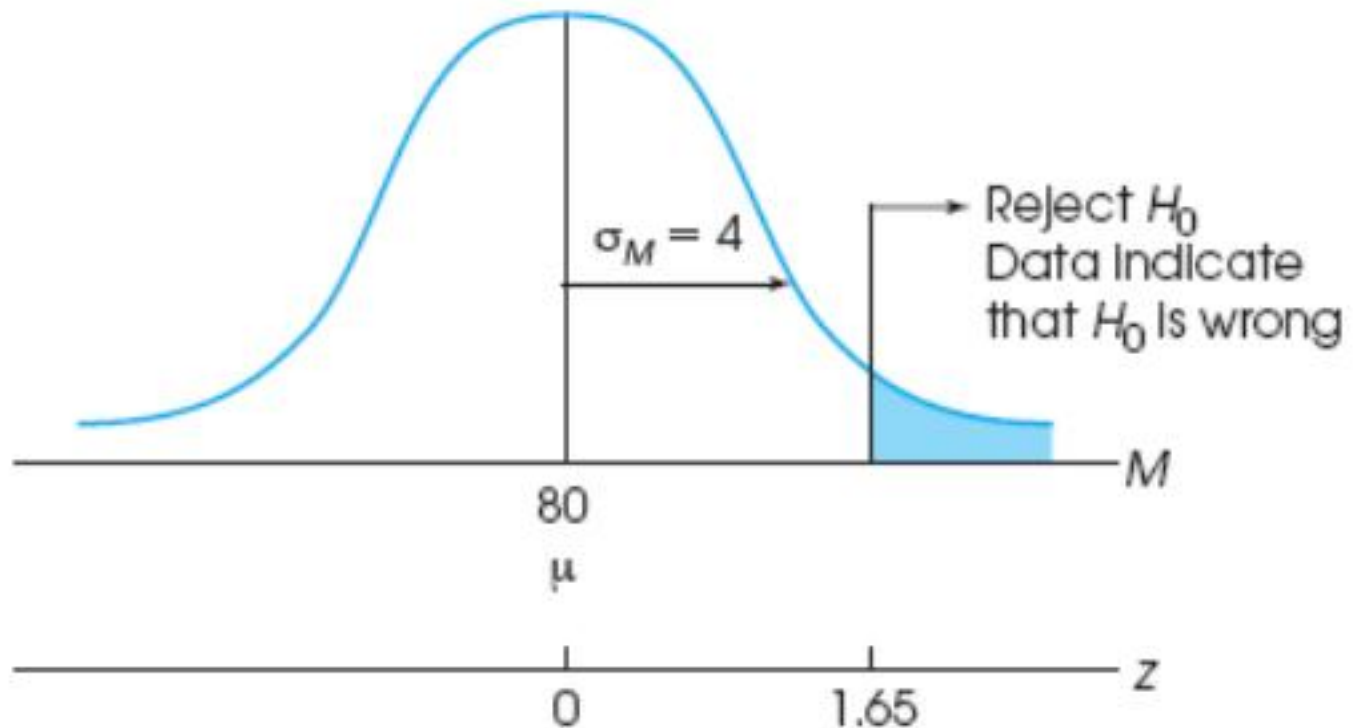
- The standard hypothesis testing procedure is called a two-tailed (non-directional) test because the critical region involves both tails to determine if the treatment increases or decreases the target behavior
- However, sometimes the researcher has a *specific prediction about the direction of the treatment*

8.4 Directional Hypothesis Tests (Continued)

- When a specific direction of the treatment effect can be predicted, it can be incorporated into the hypotheses
- In a directional (one-tailed) hypothesis test, the researcher specifies either an increase or a decrease in the population mean as a consequence of the treatment

Figure 8.7 Example 8.3

Critical Region (Directional)



One-tailed and Two-tailed Tests Compared

- One-tailed test allows rejecting H_0 with relatively small difference **provided** the difference is in the predicted direction
- Two-tailed test requires relatively large difference regardless of the direction of the difference
- In general two-tailed tests should be used unless there is a strong justification for a directional prediction

Learning Check

- A researcher is predicting that a treatment will decrease scores. If this treatment is evaluated using a directional hypothesis test, then the critical region for the test.

A

- would be entirely in the right-hand tail of the distribution

B

- would be entirely in the left-hand tail of the distribution

C

- would be divided equally between the two tails of the distribution

D

- cannot answer without knowing the value of the alpha level

Learning Check - Answer

- A researcher is predicting that a treatment will decrease scores. If this treatment is evaluated using a directional hypothesis test, then the critical region for the test.

A

• would be entirely in the right-hand tail of the distribution

B

• would be entirely in the left-hand tail of the distribution

C

• would be divided equally between the two tails of the distribution

D

• cannot answer without knowing the value of the alpha level

8.5 Hypothesis Testing Concerns: Measuring Effect Size

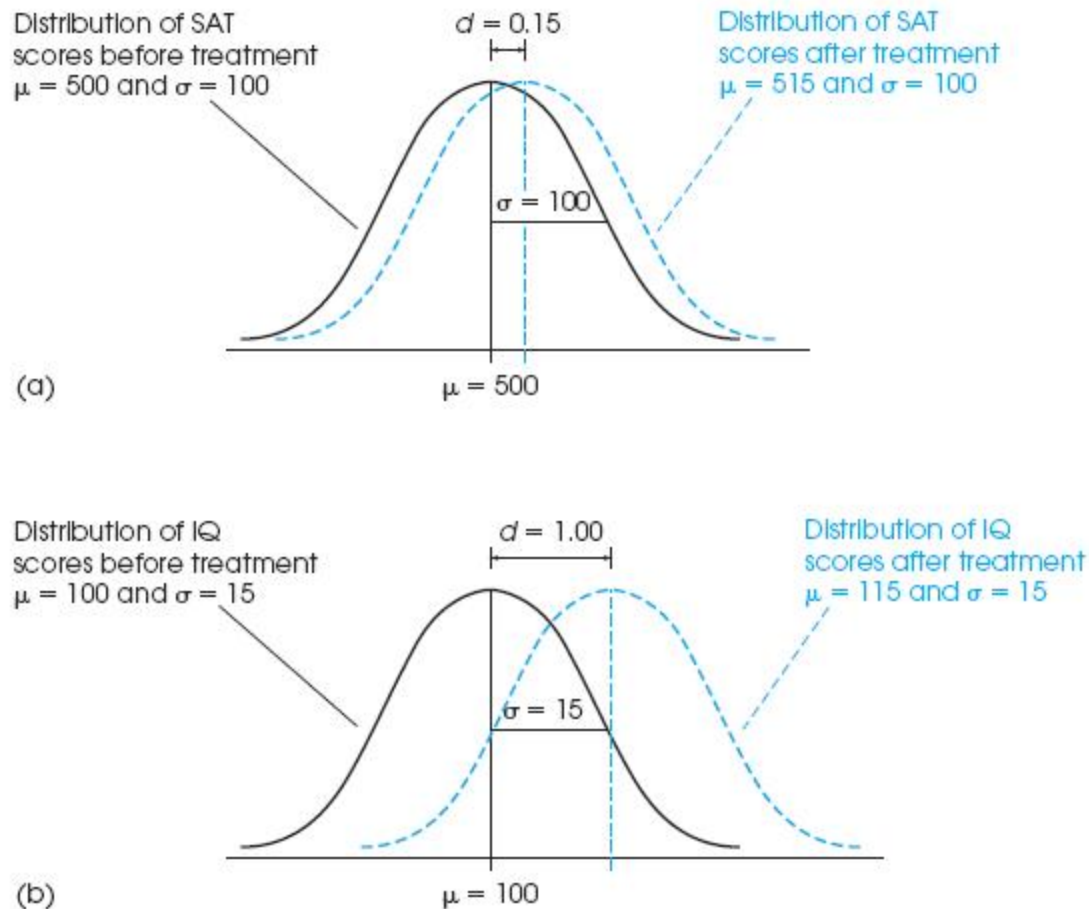
- Although commonly used, some researchers are concerned about hypothesis testing
 - Focus of test is data, not hypothesis
 - Significant effects are not always substantial
- Effect size measures the absolute magnitude of a treatment effect, independent of sample size
- Cohen's d measures effect size simply and directly in a standardized way

Cohen's d : Measure of Effect Size

$$\text{Cohen's } d = \frac{\text{mean difference}}{\text{standard deviation}} = \frac{\mu_{\text{treatment}} - \mu_{\text{no treatment}}}{\sigma}$$

Magnitude of d	Evaluation of Effect Size
$d = 0.2$	Small effect
$d = 0.5$	Medium effect
$d = 0.8$	Large effect

Figure 8.8 When is a 15-point Difference a “Large” Effect?



Learning Check

- Decide if each of the following statements is True or False.

T/F

- Increasing the sample size will also increase the effect size

T/F

- Larger differences between the sample and population mean increase effect size

Learning Check -Answers

False

- Sample size does not affect Cohen's d

True

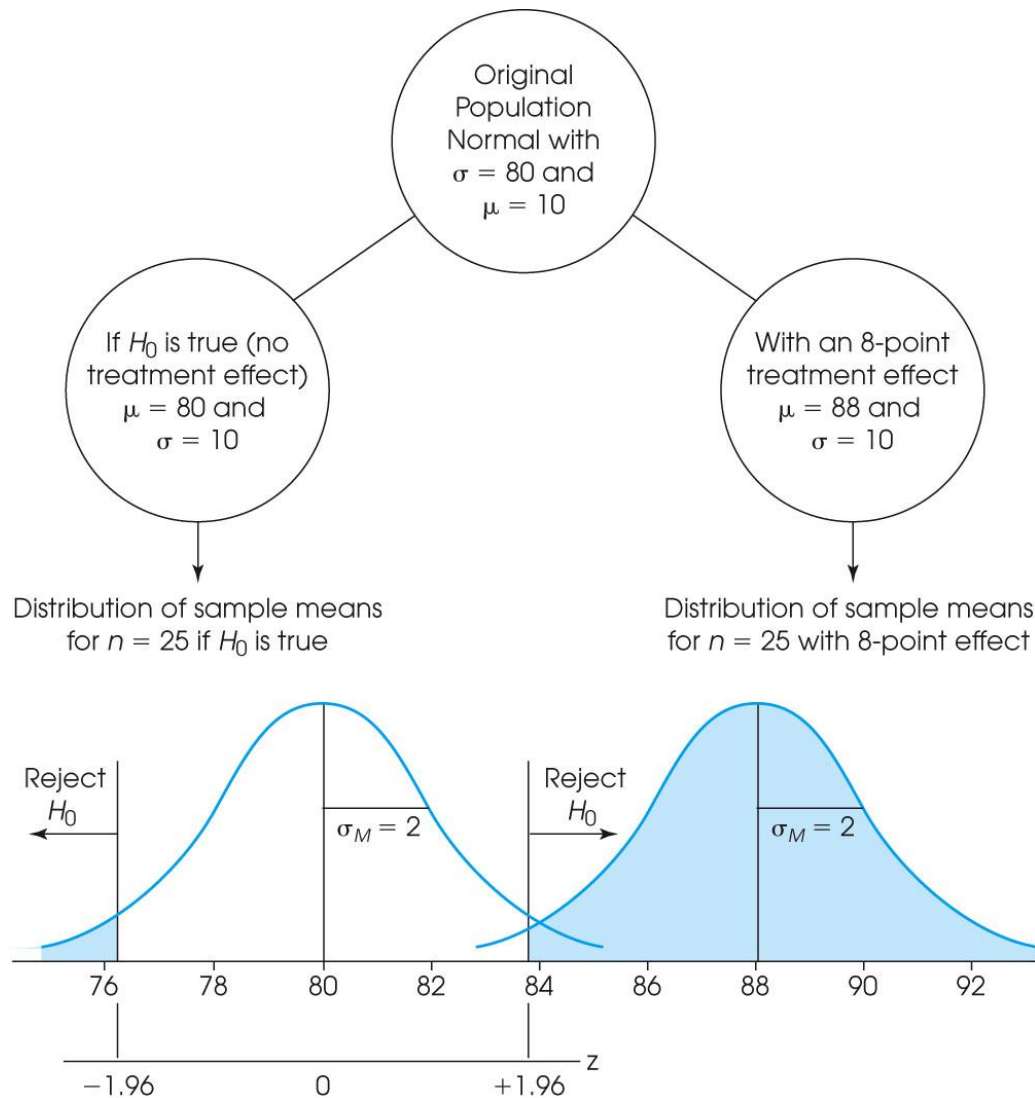
- The mean difference is in the numerator of Cohen's d

8.6 Statistical Power

- The power of a test is the probability that the test will correctly reject a false null hypothesis
 - It will detect a treatment effect if one exists
 - Power = $1 - \beta$ [where β = probability of a Type II error]
- Power usually estimated before starting study
 - Requires several assumptions about factors that influence power

Figure 8.9

Measuring Statistical Power

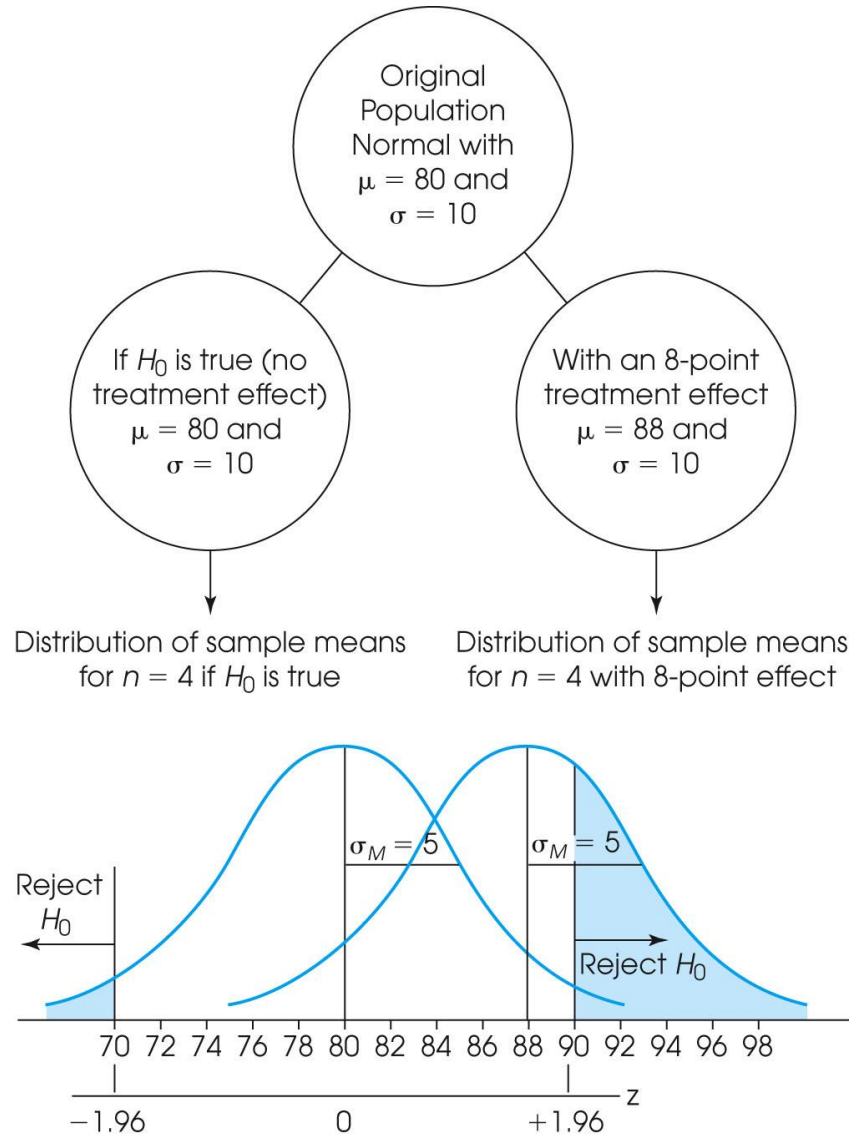


Influences on Power

- Increased Power
 - As effect size increases, power also increases
 - Larger sample sizes produce greater power
 - Using a one-tailed (directional) test increases power (relative to a two-tailed test)
- Decreased Power
 - Reducing the alpha level (making the test more stringent) reduces power
 - Using two-tailed (non-directional) test decreases power (relative to a one-tailed test)

Figure 8.10

Sample Size Affects Power



Learning Check

- The power of a statistical test is the probability of _____

A

- rejecting a true null hypothesis

B

- supporting true null hypothesis

C

- rejecting a false null hypothesis

D

- supporting a false null hypothesis

Learning Check - Answer

- The power of a statistical test is the probability of _____

A

• rejecting a true null hypothesis

B

• supporting true null hypothesis

C

• rejecting a false null hypothesis

D

• supporting a false null hypothesis

Learning Check

- Decide if each of the following statements is True or False.

T/F

- Cohen's d is used because alone, a hypothesis test does not measure the size of the treatment effect

T/F

- Lowering the alpha level from .05 to .01 will increase the power of a statistical test

Answer

True

- Differences might be significant but not of substantial size

False

- It is less likely that H_0 will be rejected with a small alpha

Chapter 9

Introduction to the t Statistic

9.1 Review Hypothesis Testing with z-Scores

- Sample mean (M) estimates (& approximates) population mean (μ)
- Standard error describes how much difference is reasonable to expect between M and μ .

• either or

•

$$\sigma_M = \frac{\sigma}{\sqrt{n}} \qquad \sigma_M = \sqrt{\frac{\sigma^2}{n}}$$

z-Score Statistic

- Use z-score statistic to quantify inferences about the population.

$$z = \frac{M - \mu}{\sigma_M} = \frac{\text{obtained difference between data and hypothesis}}{\text{standard distance between } M \text{ and } \mu}$$

- Use unit normal table to find the critical region if z-scores form a normal distribution
 - When $n \geq 30$ or
 - When the original distribution is approximately normally distributed

Problem with z-Scores

- The z-score requires more information than researchers typically have available
- Requires knowledge of the population standard deviation σ
- Researchers usually have only the sample data available

Introducing the t Statistic

- t statistic is an alternative to z
- t might be considered an “approximate” z
- Estimated standard error (s_M) is used as in place of the real standard error when the value of σ_M is unknown

Estimated standard error

- Use s^2 to estimate σ^2
- Estimated standard error:

$$\textit{estimated standard error} = s_M = \frac{s}{\sqrt{n}} \text{ or } \sqrt{\frac{s^2}{n}}$$

- Estimated standard error is used as estimate of the real standard error when the value of σ_M is unknown.

The t -Statistic

- The t -statistic uses the estimated standard error in place of σ_M

$$t = \frac{M - \mu}{s_M}$$

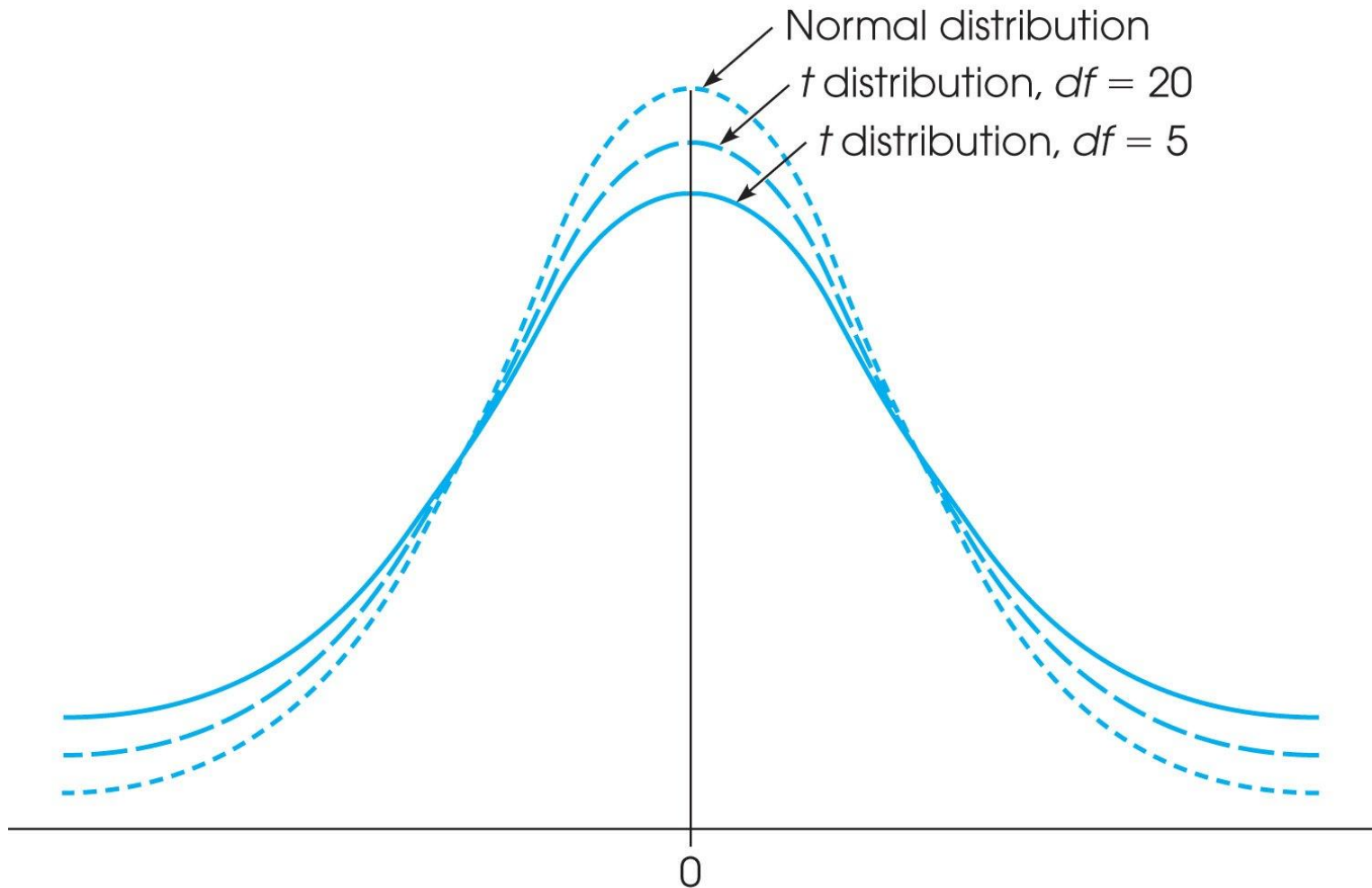
- The t statistic is used to test hypotheses about an unknown population mean μ when the value of σ is also unknown

Degrees of freedom

- Computation of sample variance requires computation of the sample mean first.
 - Only $n-1$ scores in a sample are independent
 - Researchers call $n-1$ the degrees of freedom
- Degrees of freedom
 - Noted as df
 - $df = n-1$

Figure 9.1

Distributions of the t statistic

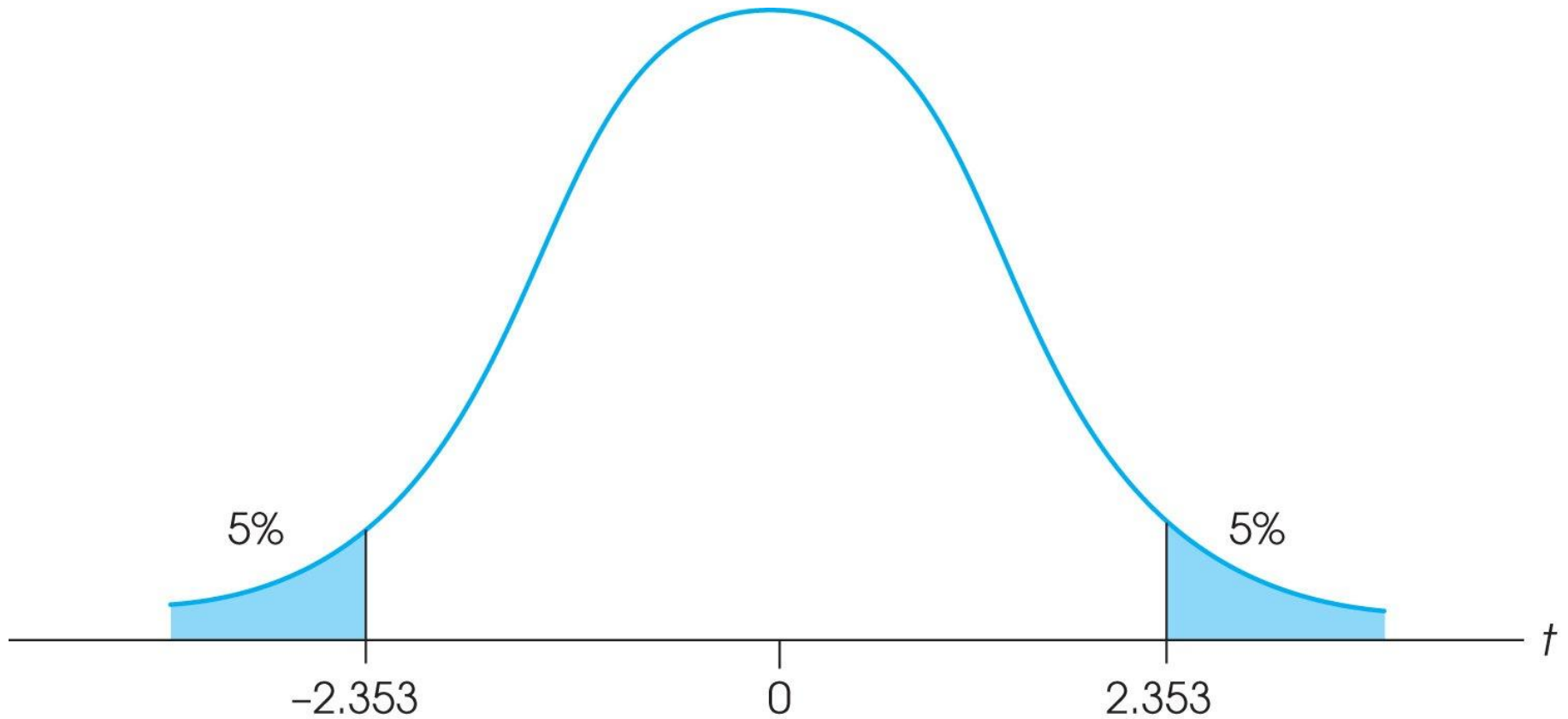


The t Distribution

- Family of distributions, one for each value of degrees of freedom
- Approximates the shape of the normal distribution
 - Flatter than the normal distribution
 - More spread out than the normal distribution
 - More variability (“fatter tails”) in t distribution
- Use Table of Values of t in place of the Unit Normal Table for hypothesis tests

Figure 9.2

The t distribution for $df=3$

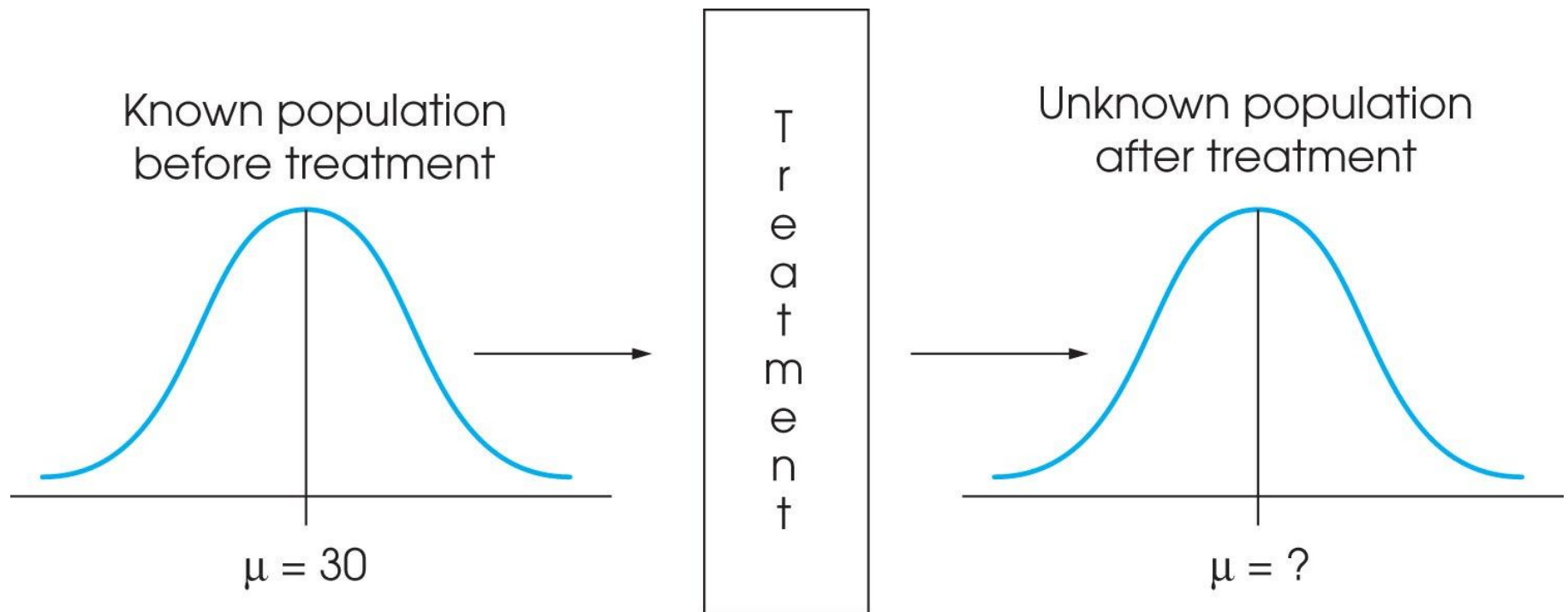


9.2 Hypothesis tests with the t statistic

- The one-sample t test statistic (assuming the Null Hypothesis is true)

$$t = \frac{\text{sample mean} - \text{population mean}}{\text{estimated standard error}} = \frac{M - \mu}{s_M}$$

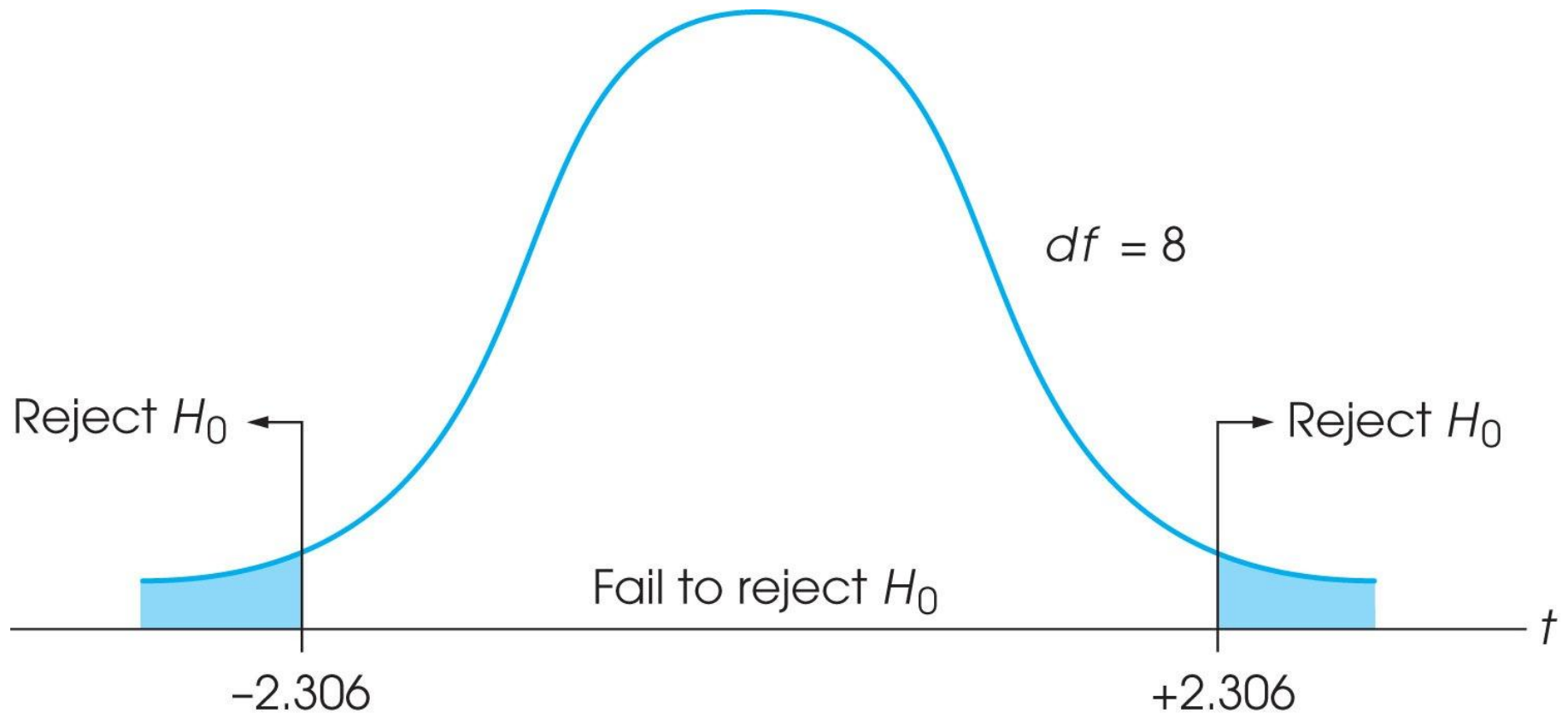
Figure 9.3 Basic experimental situation for t statistic



Hypothesis Testing: Four Steps

- State the null and alternative hypotheses and select an alpha level
- Locate the critical region using the t distribution table and df
- Calculate the t test statistic
- Make a decision regarding H_0 (null hypothesis)

Figure 9.4 Critical region in the t distribution for $\alpha = .05$ and $df = 8$



Assumptions of the t test

- Values in the sample are independent observations.
- The population sampled must be normal.
 - With large samples, this assumption can be violated without affecting the validity of the hypothesis test.

Learning Check

- When n is small (less than 30), the t distribution _____

A

- is almost identical in shape to the normal z distribution

B

- is flatter and more spread out than the normal z distribution

C

- is taller and narrower than the normal z distribution

D

- cannot be specified, making hypothesis tests impossible

Learning Check - Answer

- When n is small (less than 30), the t distribution _____

A

• is almost identical in shape to the normal z distribution

B

• is flatter and more spread out than the normal z distribution

C

• is taller and narrower than the normal z distribution

D

• cannot be specified, making hypothesis tests impossible

Learning Check

- Decide if each of the following statements is True or False

T/F

- By chance, two samples selected from the same population have the same size ($n = 36$) and the same mean ($M = 83$). That means they will also have the same t statistic.

T/F

- Compared to a z -score, a hypothesis test with a t statistic requires less information about the population

Learning Check - Answers

False

- The two t values are unlikely to be the same; variance estimates (s^2) differ between samples

True

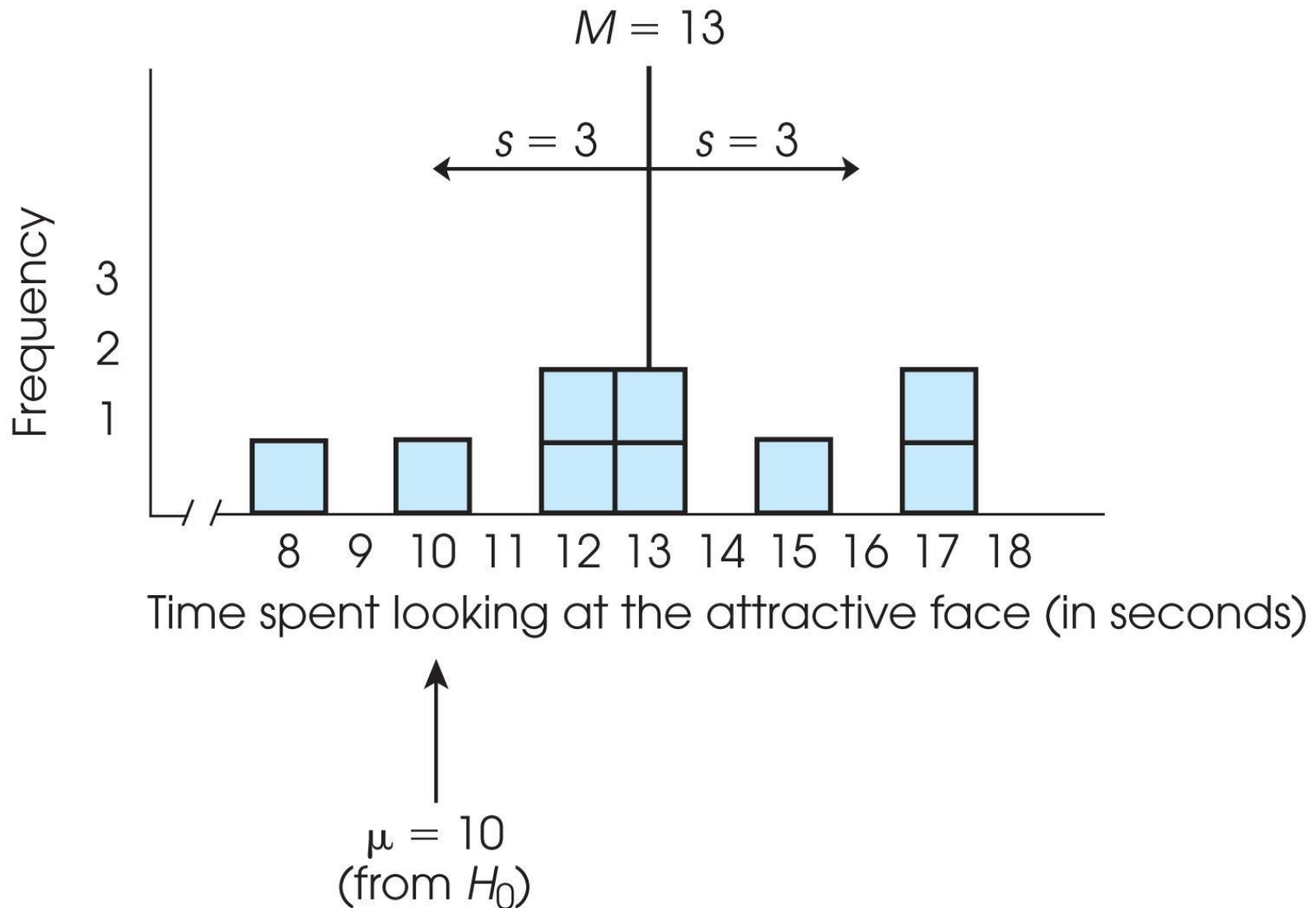
- The t statistic does not require the population standard deviation; the z -test does

9.3 Measuring Effect Size

- Hypothesis test determines whether the treatment effect is greater than chance
 - No measure of the size of the effect is included
 - A very small treatment effect can be statistically significant
- Therefore, results from a hypothesis test should be accompanied by a measure of effect size

Figure 9.5

Distribution for Examples 9.1 & 9.2



Percentage of variance explained

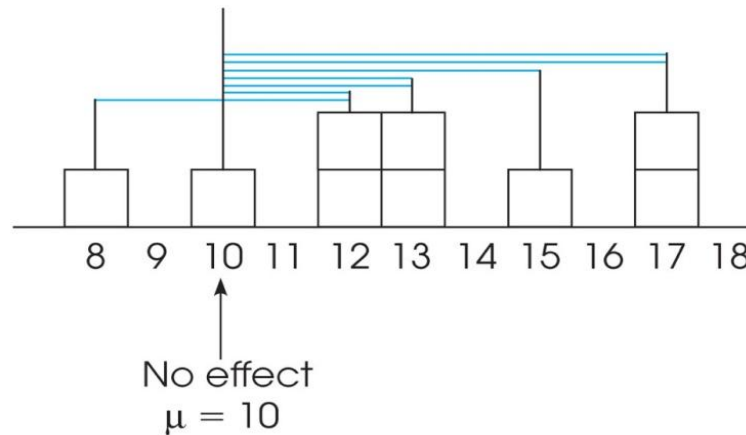
- Determining the amount of variability in scores explained by the treatment effect is an alternative method for measuring effect size.

$$r^2 = \frac{\text{variability accounted for}}{\text{total variability}} = \frac{t^2}{t^2 + df}$$

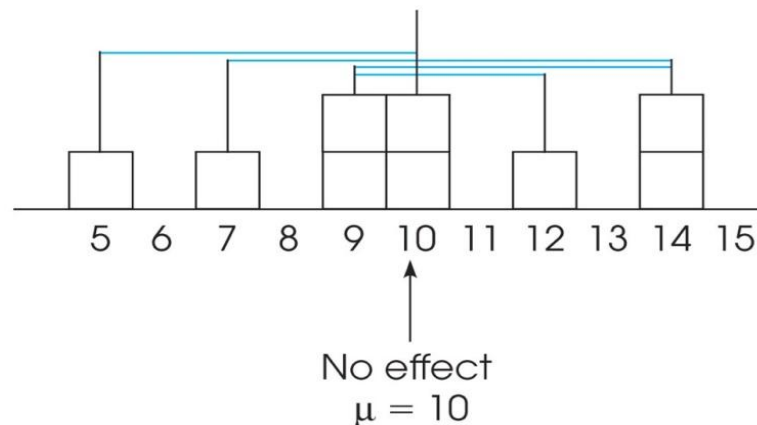
- $r^2 = 0.01$ small effect
- $r^2 = 0.09$ medium effect
- $r^2 = 0.25$ large effect

Figure 9.6 Deviations with and without the treatment effect

(a) Original scores, including the treatment effect



(b) Adjusted scores with the treatment effect removed

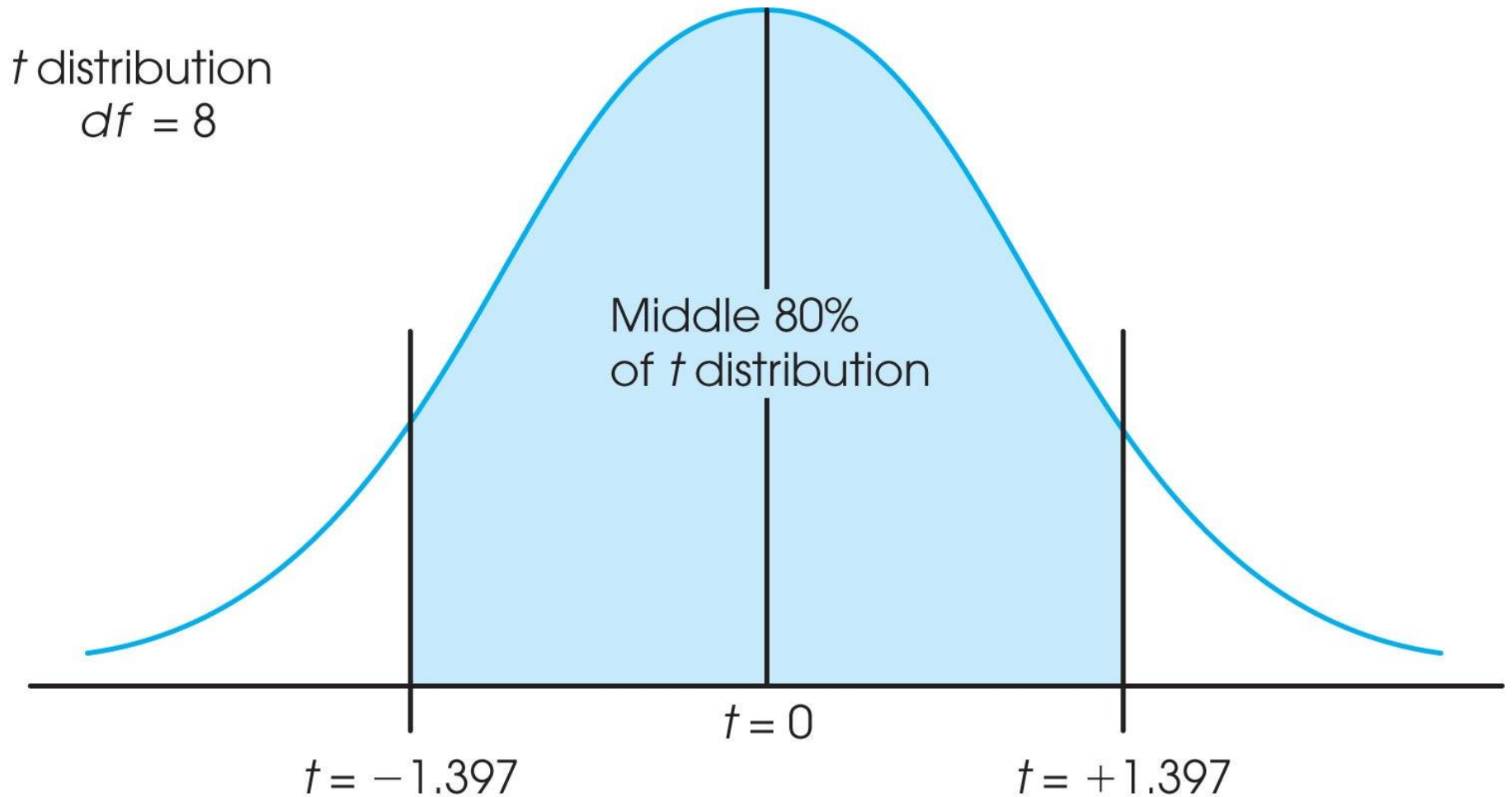


Confidence Intervals for Estimating μ

- Alternative technique for describing effect size
- Estimates μ from the sample mean (M)
- Based on the reasonable assumption that M should be “near” μ
- The interval constructed defines “near” based on the estimated standard error of the mean (s_M)
- Can confidently estimate that μ should be located in the interval

Figure 9.7

t Distribution with $df = 8$



Confidence Intervals for Estimating μ (Continued)

- Every sample mean has a corresponding t :

$$t = \frac{M - \mu}{s_M}$$

- Rearrange the equations solving for μ :

$$\mu = M \pm ts_M$$

Confidence Intervals for Estimating μ

(continued)

- In any t distribution, values pile up around $t = 0$
- For any α we know that $(1 - \alpha)$ proportion of t values fall between $\pm t$ for the appropriate df
- E.g., with $df = 9$, 90% of t values fall between ± 1.833 (from the t distribution table, $\alpha = .10$)
- Therefore we can be 90% confident that a sample mean corresponds to a t in this interval

Confidence Intervals for Estimating μ

(continued)

- For any sample mean M with s_M
- Pick the appropriate degree of confidence (80%? 90%? 95%? 99%?) 90%
- Use the t distribution table to find the value of t
(For $df = 9$ and $\alpha = .10$, $t = 1.833$)
- Solve the rearranged equation
- $\mu = M \pm 1.833(s_M)$
- Resulting interval is centered around M
- Are 90% confident that μ falls within this interval

Factors Affecting Width of Confidence Interval

- Confidence level desired
 - More confidence desired increases interval width
 - Less confidence acceptable decreases interval width
- Sample size
 - Larger sample $\rightarrow$ smaller SE $\rightarrow$ smaller interval
 - Smaller sample $\rightarrow$ larger SE $\rightarrow$ larger interval

In the Literature

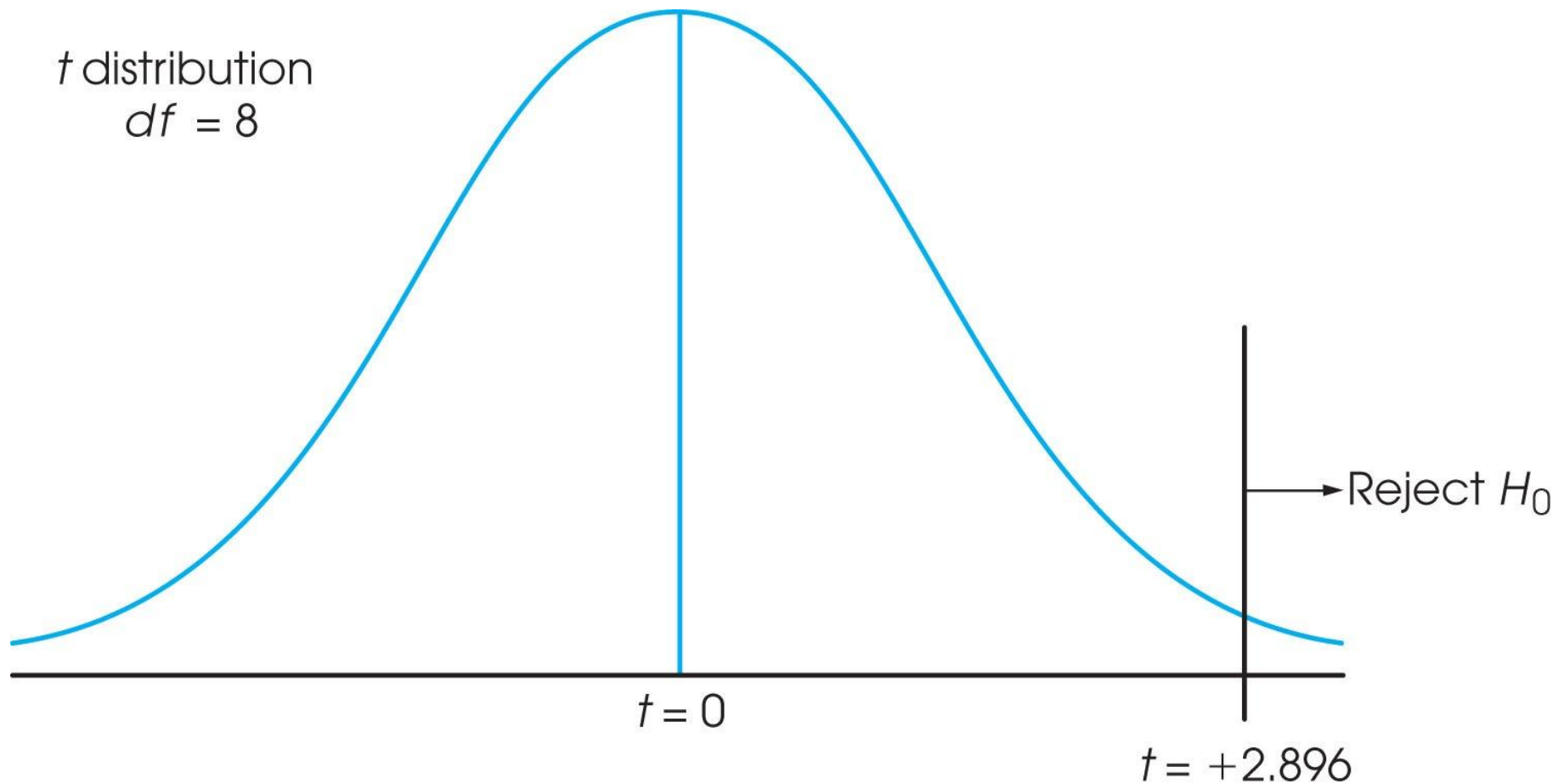
- Report whether (or not) the test was “significant”
 - “Significant” $\rightarrow H_0$ rejected
 - “Not significant” $\rightarrow$ failed to reject H_0
- Report the t statistic value including df , e.g., $t(12) = 3.65$
- Report significance level, either
 - $p < \alpha$, e.g., $p < .05$ or
 - Exact probability, e.g., $p = .023$

9.4 Directional Hypotheses and One-tailed Tests

- Non-directional (two-tailed) test is most commonly used
- However, directional test may be used for particular research situations
- Four steps of hypothesis test are carried out
 - The critical region is defined in just one tail of the t distribution.

Figure 9.8 Example 9.4

One-tailed Critical Region



Learning Check - Answer

- The results of a hypothesis test are reported as follows: $t(21) = 2.38, p < .05$. What was the statistical decision and how big was the sample?

A

- The null hypothesis was rejected using a sample of $n = 21$

B

- The null hypothesis was rejected using a sample of $n = 22$

C

- The null hypothesis was not rejected using a sample of $n = 21$

D

- The null hypothesis was not rejected using a sample of $n = 22$

Learning Check

- Decide if each of the following statements is True or False

T/F

- Sample size has a great influence on measures of effect size

T/F

- When the value of the t statistic is near 0, the null hypothesis should be rejected

Learning Check - Answers

False

- Measures of effect size are not influenced to any great extent by sample size

False

- When the value of t is near 0, the difference between M and μ is also near 0

Homework

- Chapter 8
- Q2, Q8, Q16, Q22
- Chapter 9
- Q8, Q10, Q18